

The Manhattan Project

The Manhattan Project was the US research and development program that produced the first nuclear weapons during World War II. It was led by the United States with support from the United Kingdom and Canada.

The project grew out of the Einstein-Szilárd letter of 2 August 1939, in which physicists Albert Einstein and Leó Szilárd warned President Franklin D. Roosevelt that Nazi Germany might develop nuclear weapons. The letter was delivered by Alexander Sachs.

Major General Leslie Groves of the US Army Corps of Engineers was appointed director in September 1942. He appointed physicist J. Robert Oppenheimer to lead the secret weapons laboratory at Los Alamos, New Mexico.

The first nuclear device, codenamed "the Gadget", was detonated in the Trinity test on 16 July 1945 at the Alamogordo Bombing Range in New Mexico. The "Fat Man" plutonium bomb was dropped on Nagasaki on 9 August 1945; the "Little Boy" uranium bomb was dropped on Hiroshima on 6 August 1945.

Major sites included the Hanford Site in Washington state (plutonium production, led by Glenn Seaborg), the Clinton Engineer Works at Oak Ridge, Tennessee (uranium enrichment), and the Metallurgical Laboratory at the University of Chicago (led by Enrico Fermi, who built the first nuclear reactor, Chicago Pile-1, on 2 December 1942).